

# Amir Mirzai Golpayegani

📍 Calgary, AB, Canada

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## Summary

M.Sc. Computer Science graduate with experience building Python/C++ systems for machine learning, backend APIs, data pipelines, and large-scale data processing. Skilled in PyTorch, Scikit-learn, REST APIs, SQL, Docker, Git, and reproducible experimentation, with applied research experience in deep learning, AI workflow automation, geospatial data infrastructure, and production-minded software development.

## Education

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| <b>M.Sc. in Computer Science</b><br><i>University of Calgary</i>               | GPA: <b>3.94/4.0</b> | Sep 2022 – Feb 2026<br><i>Calgary, Canada</i> |
| <b>B.Sc. in Computer Engineering</b><br><i>Sharif University of Technology</i> | GPA: <b>3.69/4.0</b> | Sep 2016 – Feb 2021<br><i>Tehran, Iran</i>    |

## Technical Skills

**Programming:** Python, C++, JavaScript, SQL

**ML / AI:** PyTorch, Scikit-learn, TensorFlow, CNNs, model training, evaluation, LLM workflows, AI agents

**Data / Pipelines:** NumPy, Pandas, SciPy, Google Earth Engine API, GDAL, Rasterio, SNAP, DGGs

**Backend / Engineering:** Django, Node.js, REST APIs, PostgreSQL, Docker, Git, Linux, AWS, GitHub Actions

**Visualization / Automation:** Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

## Work Experience

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| <b>Back-End Developer</b><br><i>Asr Gooyesh Pardaz</i> | Jun 2020 – Jul 2022<br><i>Tehran, Iran</i> |
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- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

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| <b>Research Software Developer</b><br><i>BigGeo / Mitacs, University of Calgary</i> | Sep 2022 – Feb 2026<br><i>Calgary, Canada</i> |
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- Built C++/Python software for scalable processing, storage, and retrieval of Earth observation data.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGs-based encoding pipelines to convert satellite imagery into structured tensor files.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Reduced storage by 38% for scaled Canada-wide data with 1,112 GeoTIFF tiles and 16K+ DGGs tensors.

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| <b>Deep Learning Teaching Assistant</b><br><i>Neuromatch Academy</i> | Jul 2026<br><i>Remote, United States</i> |
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- Mentored an international pod of ~15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

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| <b>Teaching Assistant</b><br><i>Department of Computer Science, University of Calgary</i> | Dec 2022 – Apr 2025<br><i>Calgary, Canada</i> |
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- **Developing Big Data Applications:** Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale:** Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems:** Assisted +75 students with SQL, database design, course projects, grading, and exam support.

## Research and Technical Projects

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|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Agentic Job Search Automation Platform</b><br><i>OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API</i> | May 2026 – Present |
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- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.
- Designed LLM-based workflows for resume/job matching, skill-gap analysis, and application decision support.

- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.
  - Prototyped the initial workflow in n8n before migrating toward a more flexible OpenClaw-based agent system.
- DEM Super-Resolution Framework

May 2024 – Feb 2026

Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization

  - Trained deep ResNet models for 4× DEM super-resolution, reconstructing 30m elevation data from 120m inputs.
  - Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and collect mountainous terrain data worldwide.
  - Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors.
  - Developed a Polyscope-based 3D visualization tool to render DEM tiles from multiple angles and compare reconstruction quality in 3D.
- Spatiotemporal Earth Observation Data System Using DGGs

Sep 2022 – Feb 2026

First-author publication | M.Sc. thesis | C++, Python, DGGs, Wavelets, B-splines, Time-Series

  - Designed a DGGs-based framework for real-time multiresolution management of satellite imagery.
  - Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.
  - Retrieved 105,489 DGGs cells per time-series query in 13.9s, compared with about 220s using global retrieval.

Publication

Amir Mirzai Golpayegani, Mahmudul Hasan, Faramarz F. Samavati. *Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGs*. Remote Sensing, 2025. [doi:10.3390/rs17040570](https://doi.org/10.3390/rs17040570)

Certificates

Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford University, Coursera    Oct 2024  
Applied Machine Learning in Python, University of Michigan, Coursera    Jul 2020

Languages

English: Full Professional Proficiency (TOEFL 115/120)  
Persian: Native Proficiency

Achievement

Teaching Assistant Excellence Laureate    2024/2025  
Department of Computer Science, University of Calgary — awarded for outstanding teaching support ([link](#))  
Ranked 17th Nationwide in Iran University Entrance Exam (Konkour)    May 2016  
Top 0.001% among all participants